

Assignment 2

CFM

This assignment builds upon the dataset collected in Assignment 1. Students must use the stock data gathered earlier for their selected index.

1. Data Verification and Cleaning

- a) Check for missing values
- b) Check for duplicate records
- c) Ensure data is properly ordered by date
- d) Verify price adjustments for splits/dividends
- e) Check continuity of trading days

2. Exploratory Data Analysis (EDA)

- a) Plot price time series for all selected stocks
- b) Compute daily returns
- c) Plot return time series
- d) Compute descriptive statistics (mean, standard deviation, skewness, kurtosis)

Stock	Mean Return	Std Dev	Skewness	Kurtosis

4. Normality tests

- a) Plot histogram of returns
- b) Create Q-Q plot
- c) Perform Jarque-Bera test
- d) Perform Shapiro-Wilk test

5. Correlation Analysis

- a) Compute correlation matrix
- b) Create correlation heatmap
- c) Identify most and least correlated stock pairs

6. Random Portfolio Generation

Select any two stocks from your dataset and generate random portfolios.

- a) Generate at least 5000 random portfolios
- b) Use random weights for each portfolio
- c) Compute portfolio return and risk (standard deviation)
- d) Plot risk-return scatter diagram

7. Markowitz Portfolio Optimization

- a) Compute expected returns
- b) Compute covariance matrix
- c) Solve minimum variance portfolio
- d) Generate portfolios for different target returns
- e) Plot random portfolios of last questions along with the efficient frontier curve

- f) Clearly indicate minimum variance portfolio

9. Observations

- a) Comment on distribution of stock returns
- b) Explain shape of feasible portfolio region
- c) Interpret efficient frontier
- d) Explain minimum variance portfolio

Deliverables

Python Notebook containing

- a) all code
- b) Charts and visualizations
- c) Tables of descriptive statistics
- d) Risk-return plot of random portfolios
- e) Efficient frontier plot
- f) Short report (2–4 pages) explaining observations